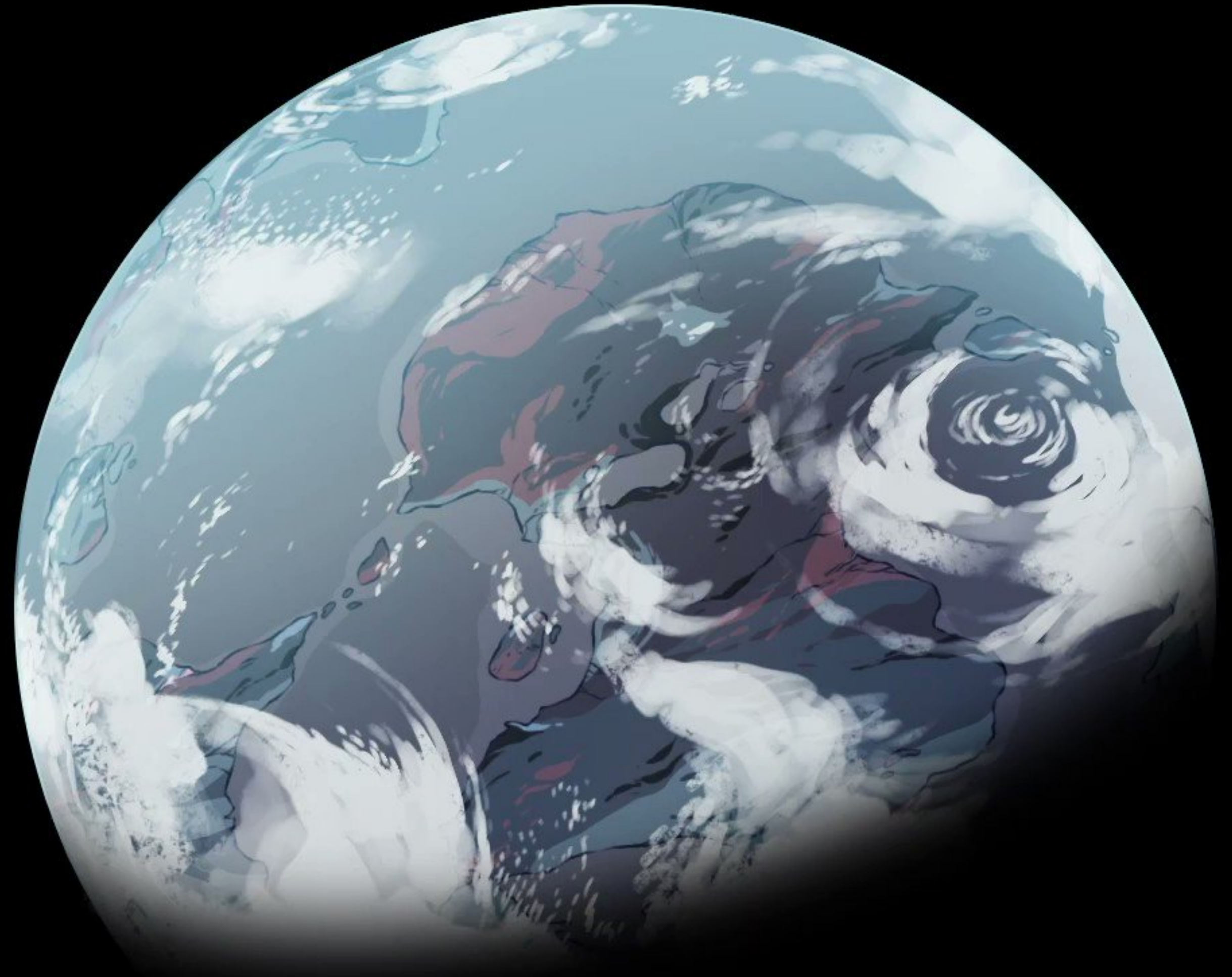


Science & Science Fiction

33-120 | Prof. Katie Breivik



- ▶ Exploration paper 3 is available
 - ▶ due Wednesday, March 19 @ midnight
- ▶ Today: Machine Learning, Artificial Consciousness

Imagining a Realistic Scenario for First Contact

Now that we have covered the search for extraterrestrial intelligence in class, your group is equipped to imagine a realistic scenario for *first contact* with extraterrestrial life. While we don't know exactly how first contact would play out since we are pretty sure that it hasn't happened (..yet) your group should work to maintain scientific plausibility throughout your scenario.

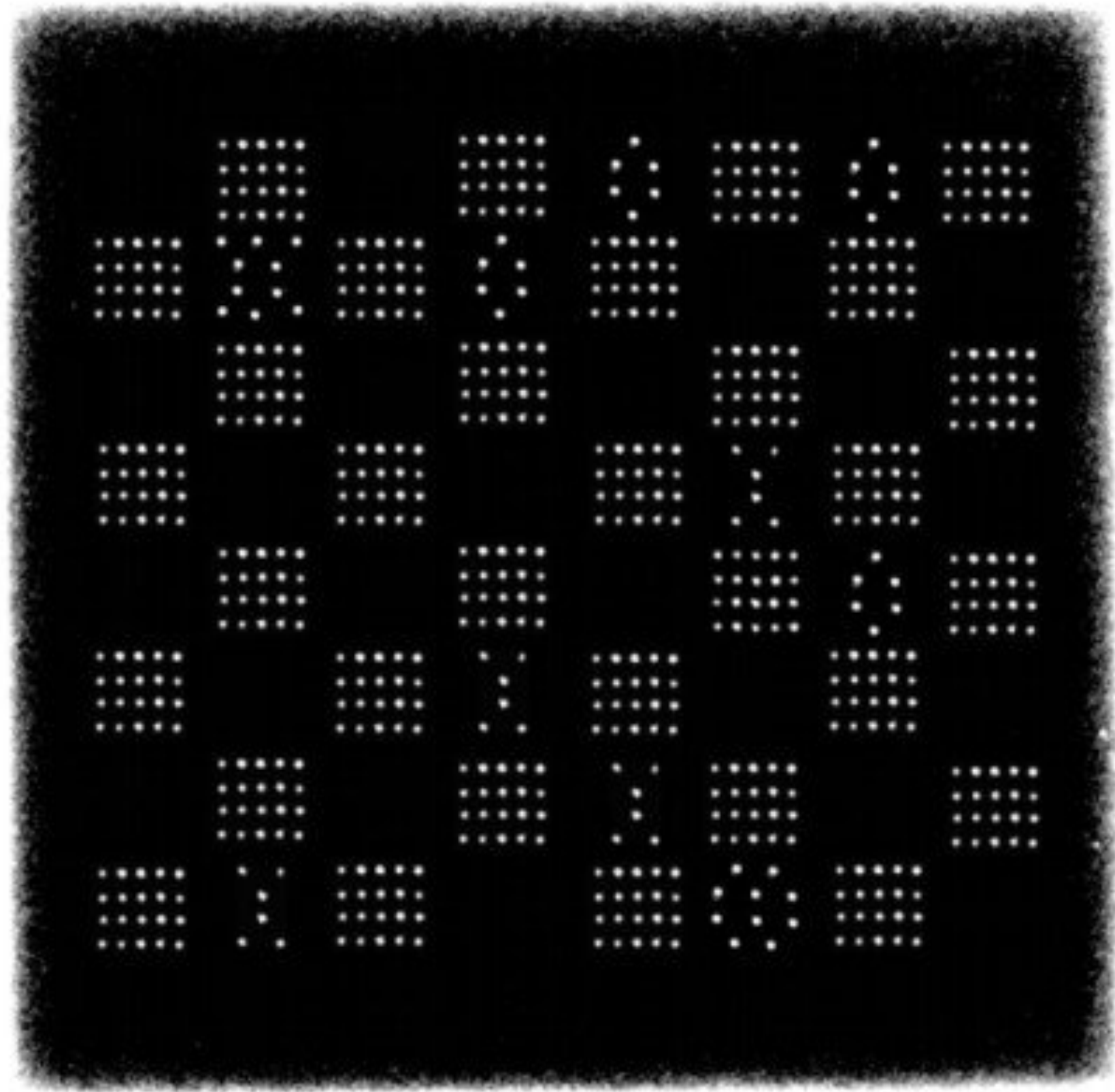
As a starting point, you should assume that the contact is *not an invasion*. However, it is reasonable to assume that any alien intentions are unclear until the scenario plays out. Throughout the project, you should consider realistic factors like communication barriers (both physical and social) as well as social response (both general public as well as administrative/governmental).

SETI Team Project

Teams are assigned on Canvas
— find your team and decide on
a time/place to meet.

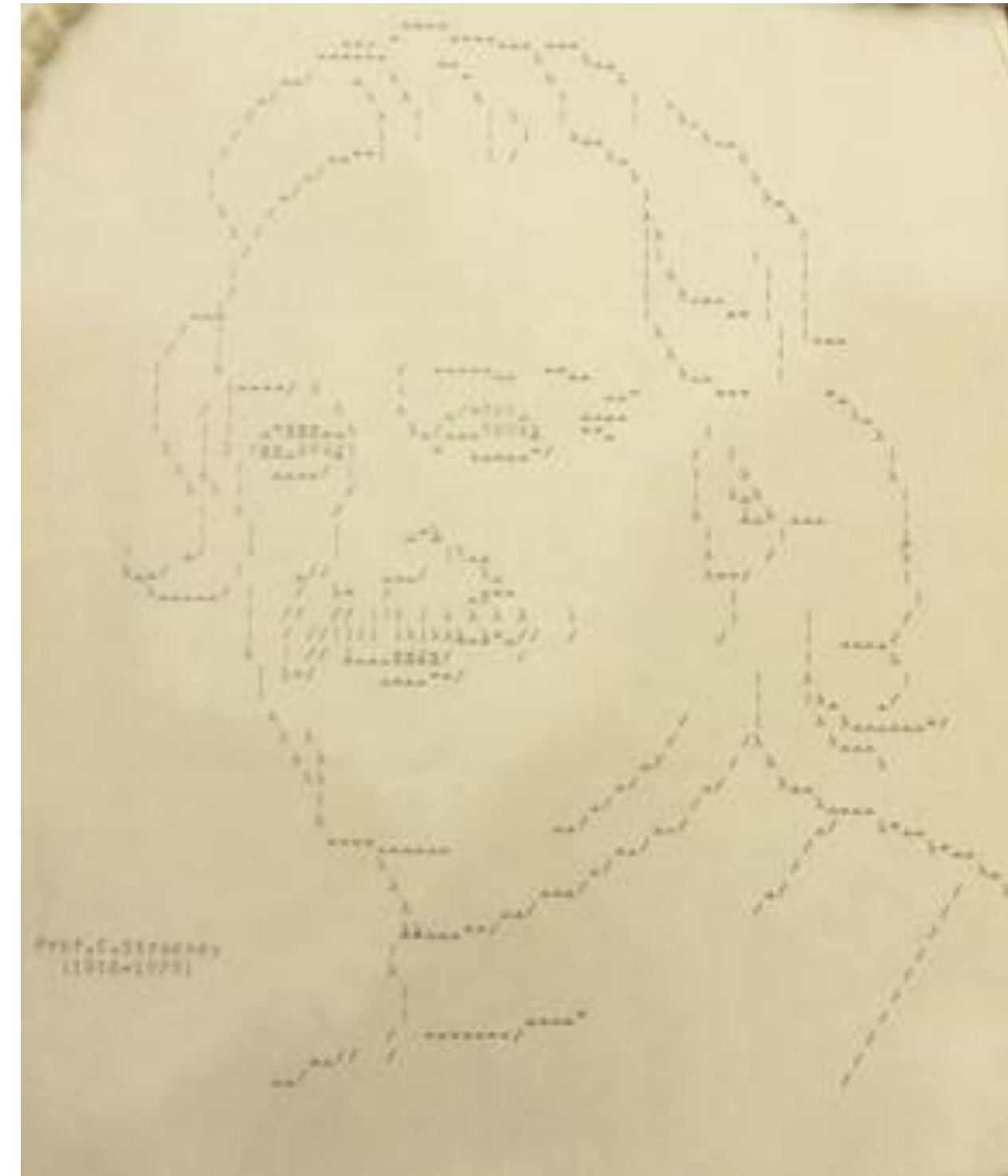
Class time on March 28 has
been reserved as a time that is
guaranteed to work for
everyone!

Machine Learning



It all started with Checkers

Christopher Strachey



Developed first Checkers program in summer 1951, but it took until summer 1952 to get the program running at a normal speed and without bugs

Machine Learning

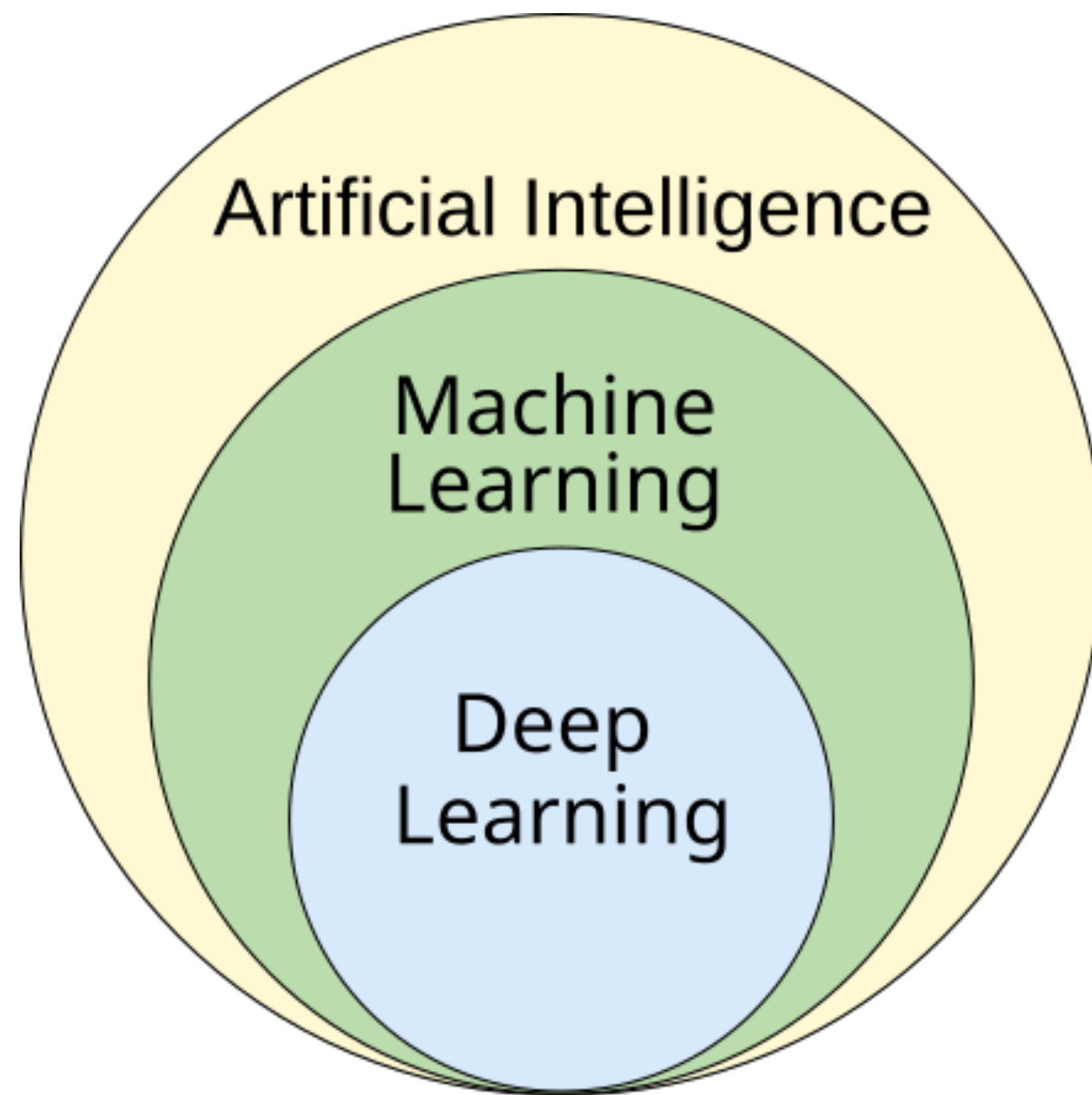


Coined by
Arthur Lee Samuel
in 1959

Wrote the Samuel Checkers-
playing Program in 1956

Rather than computing all possible outcomes, the program implements 'alpha-beta pruning' which considers moves based on the current state of the board and leaves behind all choices that are not likely to be as successful as the remaining choices — much more efficient than computing all possible outcomes!

Machine Learning



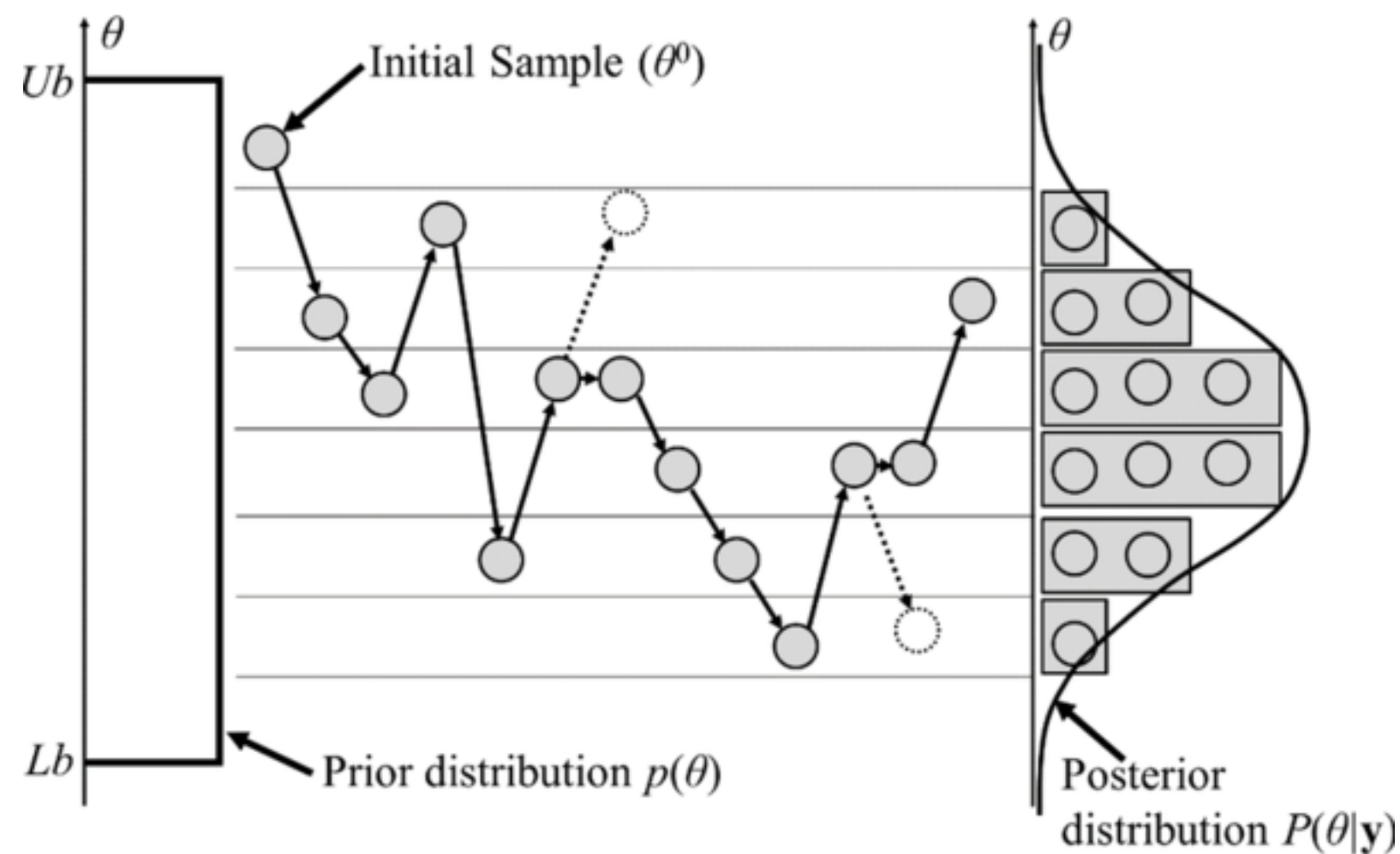
A sub-discipline of AI —
but not just deep learning

Two objectives:

1. Classify data based on already developed models
2. Make predictions for future outcomes based on those models

Machine Learning that is not Deep Learning: Statistics

Supervised learning: we know what goes in and out of the model



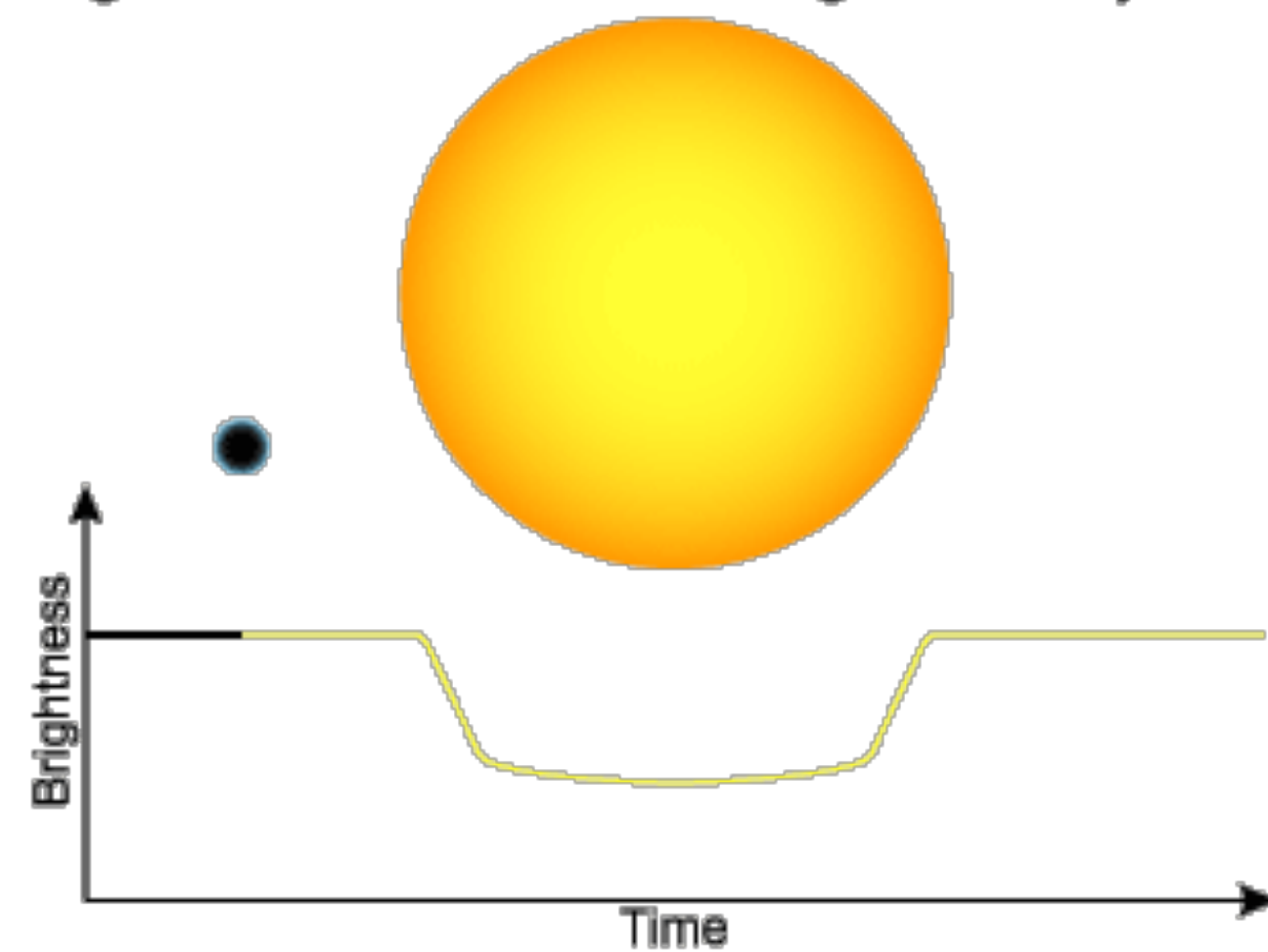
Metropolis-Hastings algorithm:
An optimized guess and check

- 1) start with a 'prior distribution' which defines your 'prior knowledge' of the model parameters
- 2) sample a parameter set from the prior
- 3) Evaluate the model with the sampled parameter set and compare to data
- 4) If previous sample fits the data better, add that sample to a list; if not add the newest sample to the list
- 5) repeat!

Machine Learning that is not Deep Learning: Statistics

Fitting exoplanet models to transit data

Light Curve of a Star During Planetary Transit

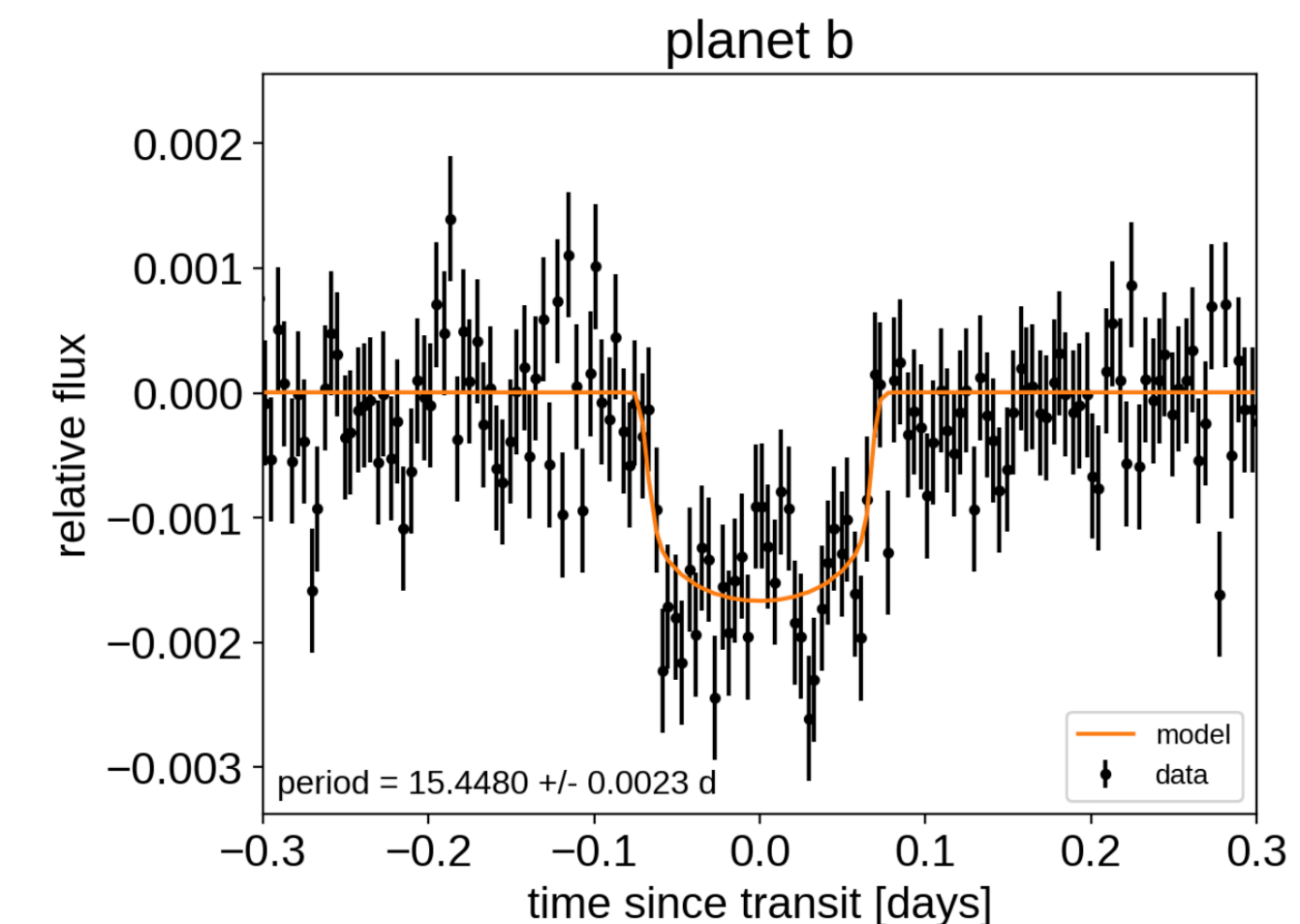


This process is described by a Keplerian orbit

Model parameters:

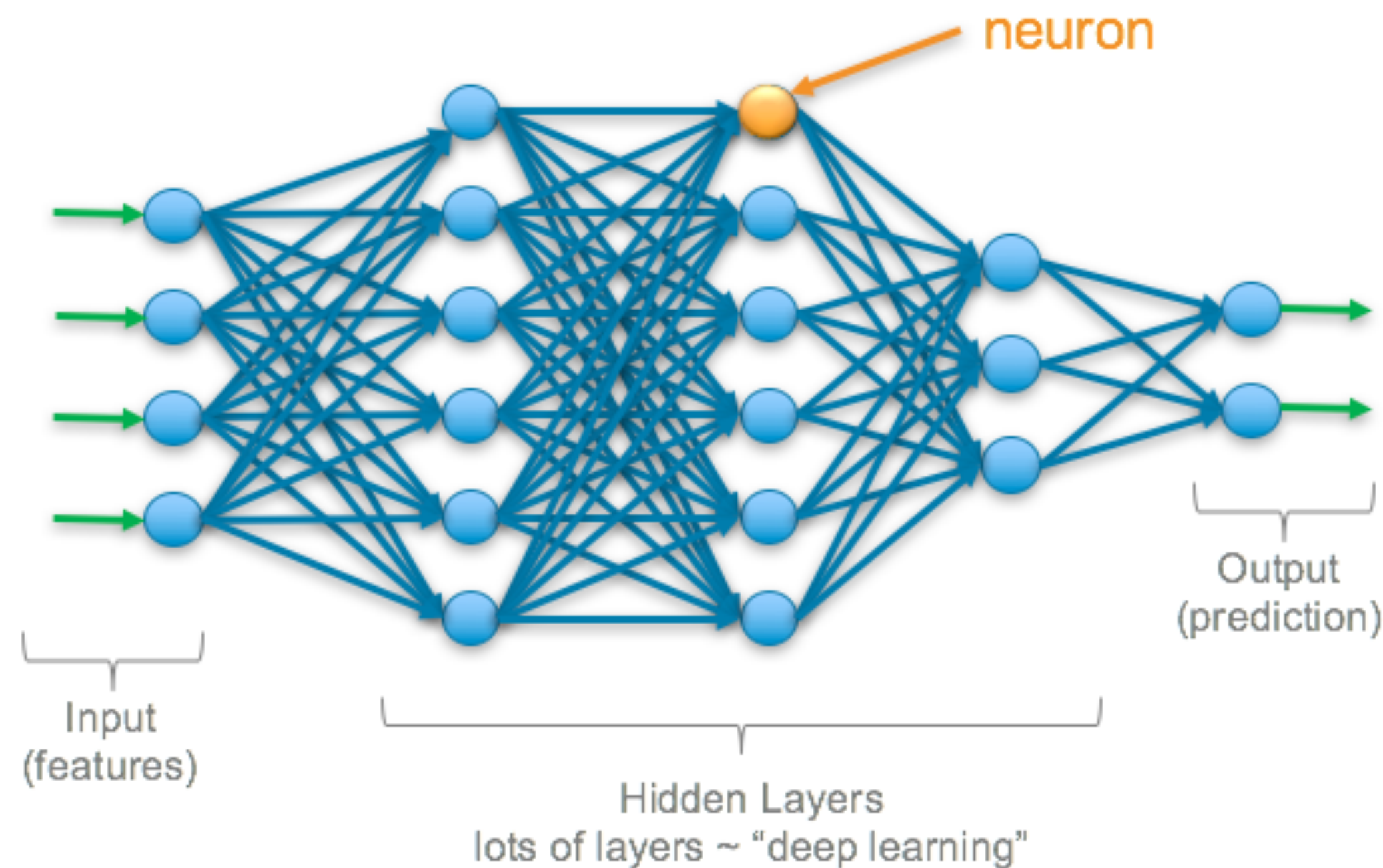
M_{star} , M_{planet} , P_{orb} , e , L_{star} , R_{star} , R_{planet}

Sample over different parameters
to fit observed light curve



The one that fits the best tells us
the properties of the system!

Machine Learning that is Deep Learning



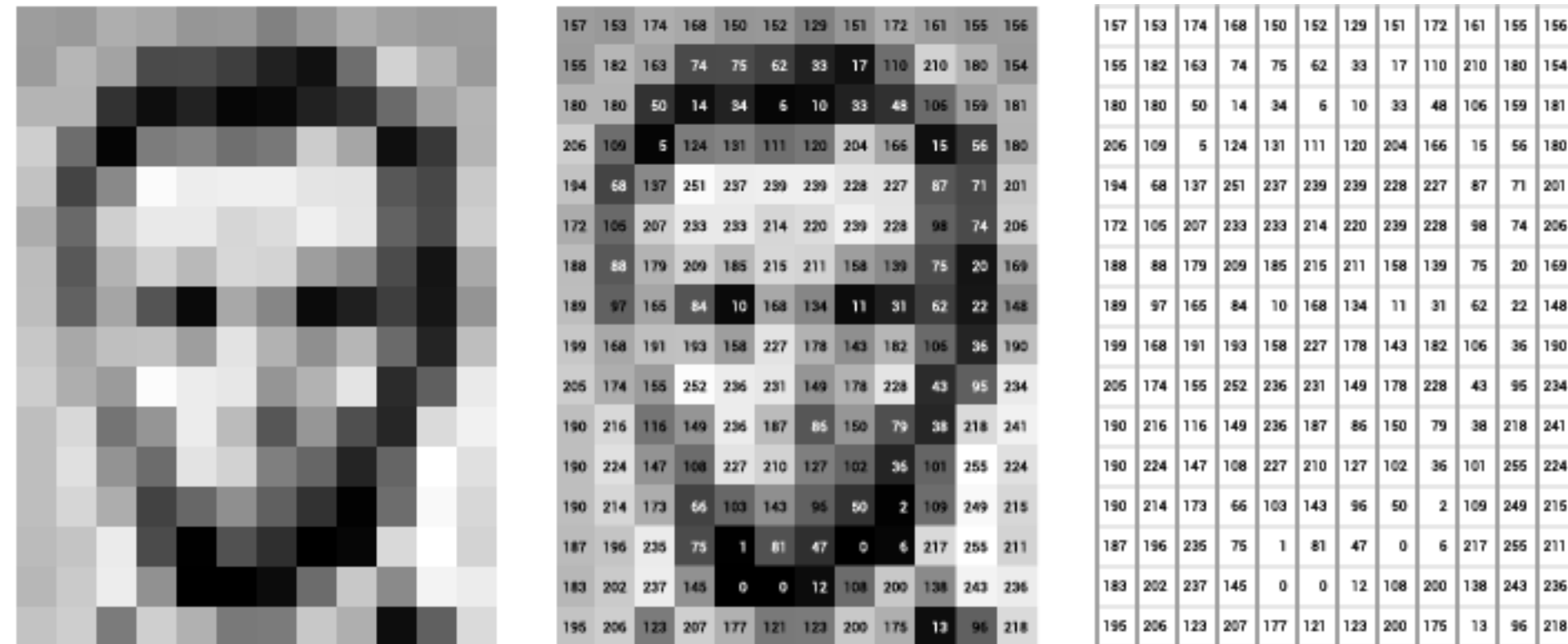
Feature extraction
happens automatically

—
no model required

—
A **lot** of training required

It is often difficult to ***interpret*** the results of deep learning algorithms

examples of deep learning algorithms



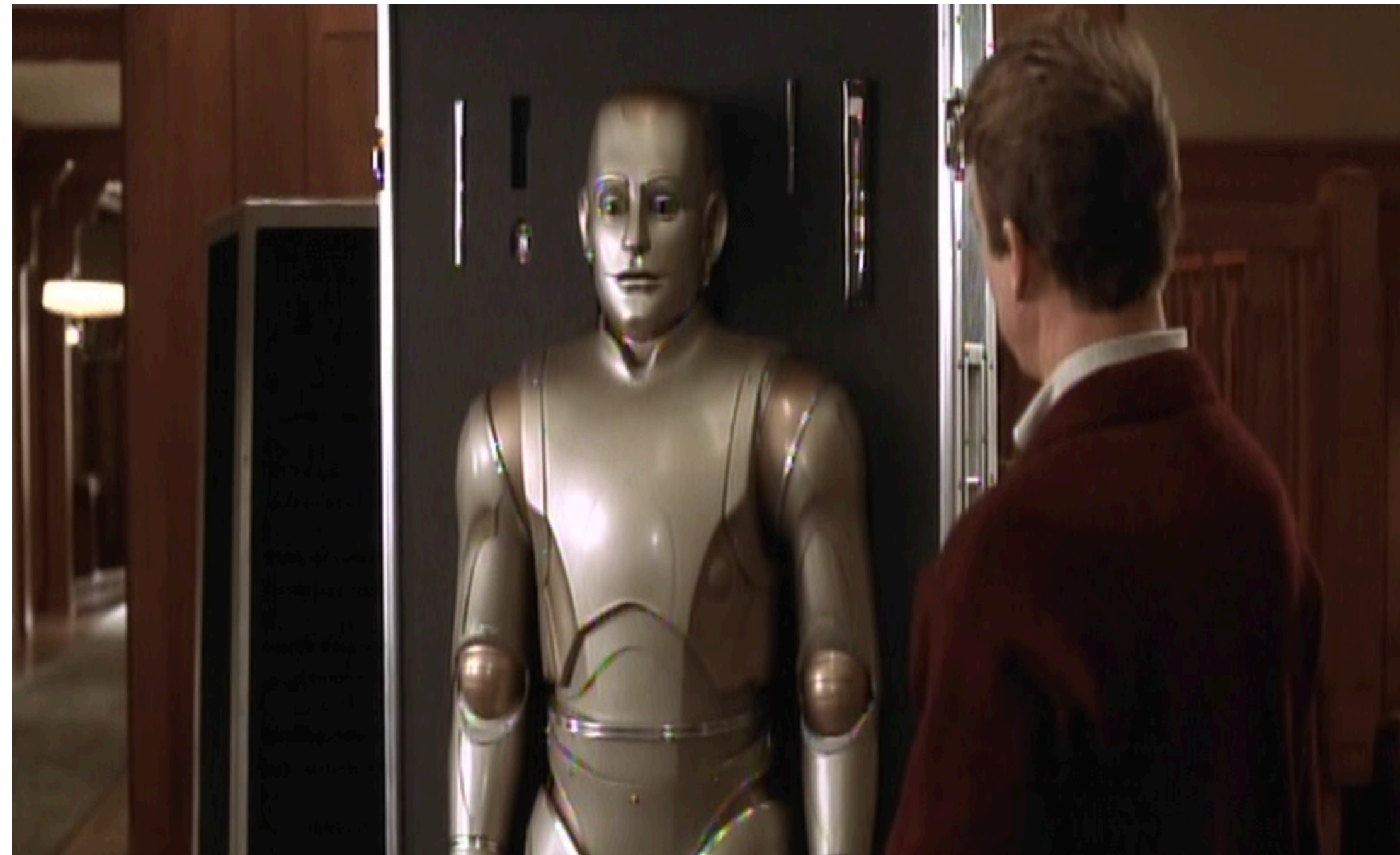
computer vision

- Map images to tabular data that can be analyzed and classified by a computer
- Used heavily in self-driving car technology
- Images are often compressed to lower dimensions because of computational cost



large language models

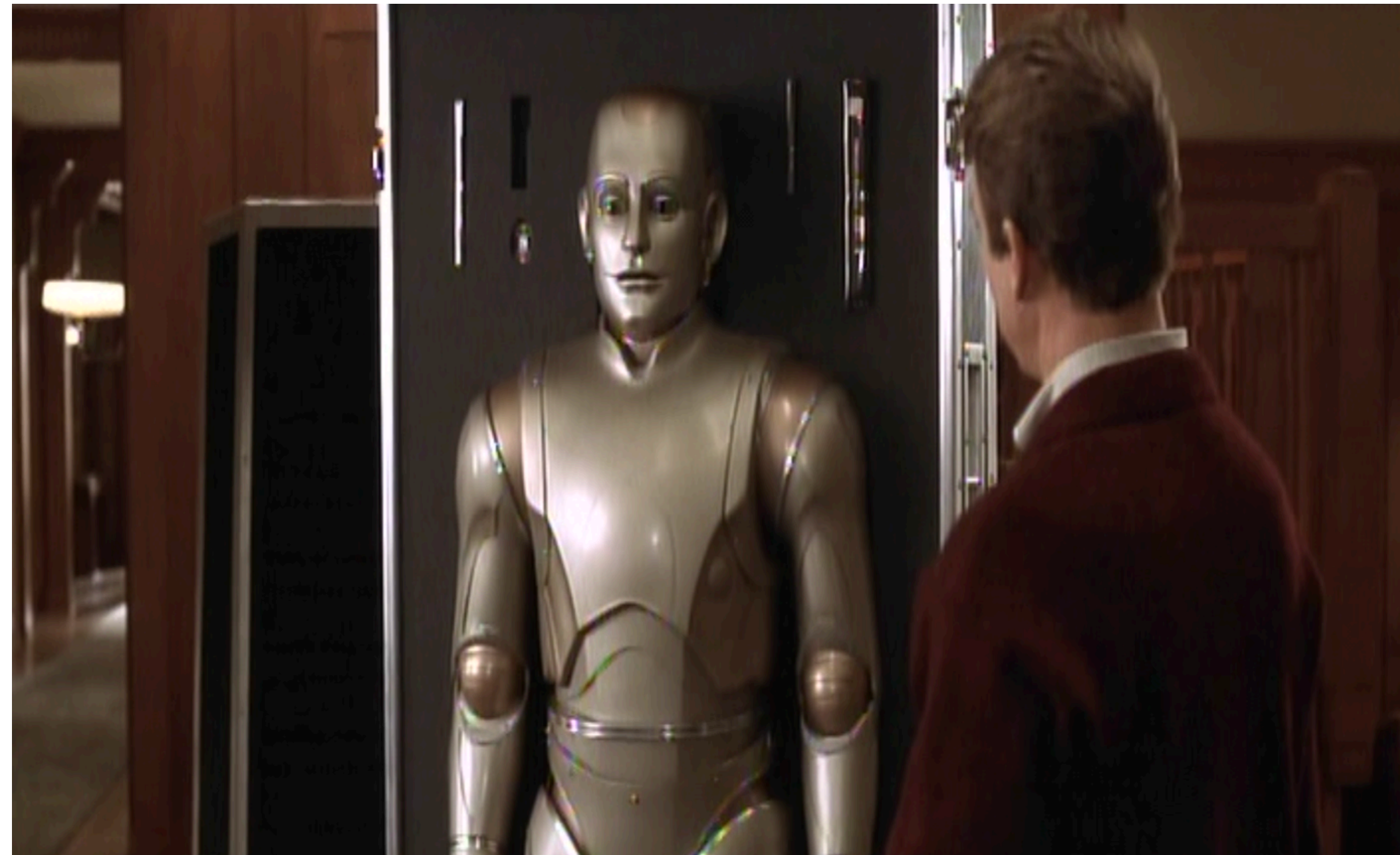
- Rely on *enormous* data sets to train autoregressive models that use past data sets to predict future values
- Extremely expensive training:
GPT-4 cost \$100 million



Bicentennial Man

Directed:
Chris Columbus

Columbia Pictures
(1999)



How does Andrew learn?

Statistical machine
learning or
deep learning?

How realistic is the depiction of Andrew's learning?



Chappie

Directed:
Neill Blomkamp

Sony Pictures
(2015)



How does Chappie learn?

Statistical machine learning or deep learning?

How realistic is the depiction of Chappie's learning?

There is no empirical definition of consciousness

Qualia: instances of *subjective* conscious experiences

First termed by Clarence Irving Lewis in 1929

Pain from a headache

Taste of food

Redness of a sunset

Some argue that a robot could be determined to be conscious if it is able to make judgements based on qualia

Scavenger's Reign

“The Dream”

Written:

Joseph Bennett
Charles Huettnner
Sean Buckelew

Titmouse, Inc | Green
Street Pictures (2023)



Asimov's Laws of Robotics:

"Handwritten" A.D. 1042
These are not official
laws in any way!

► A robot must not harm a human being, or, through inaction, allow a human being to come to harm.

► A robot must obey the orders of a human being, except where such orders would conflict with the First Law.

► A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

- ▶ Exploration paper 3 due Wednesday!
- ▶ Next time: Begin “Are we alone in the Universe?”